Operational KPI & Marketing Metric Board

Technical Project Documentation & Google Sheets Implementation Blueprint

PROJECT DOMAIN	APPLICATION	AESTHETIC	ARCHITECTURE
Marketing Analytics	**Google Sheets**	**Clean Minimalist (#FFFFFF)**	**4-Sheet Relational Model**

1. Executive Summary & Core Objectives

This technical report outlines the end-to-end design, data engineering, calculation modeling, and interactive dashboarding developed for the enterprise analytics project. The workbook establishes an executive minimalist operational command center in Google Sheets, engineered to ingest 100 multi-channel marketing transaction records, compute automated aggregations, and deliver executive intelligence through dynamic KPI blocks and live slicers.

2. Complete 4-Sheet Workbook Architecture

The solution is organized into four purpose-built tabs to maintain strict separation of ingestion, aggregation, presentation, and metadata:

- `Raw_Data`: Ingestion layer housing 100 enterprise marketing campaign transaction records with unique campaign identifiers, timestamps, channels, impressions, clicks, spend, conversions, and revenue.
- `Pivot_Engine`: Calculation backend utilizing error-handled aggregation formulas (`SUMIF`) to synthesize channel-level performance metrics dynamically.
- `Dashboard`: Executive command center featuring top-line summary cards, linked data tables, custom data visualizations, and interactive slicers.
- `Lookup_Reference`: Master repository storing channel target ROAS benchmarks, monthly budget caps, and assigned campaign managers.

3. Data Engineering & Calculation Formulas

All quantitative metrics are driven by dynamic, audit-ready formulas across sheets to ensure real-time responsiveness when filtering:

```
      # 1. Row-Level ROAS Calculation (Raw_Data Sheet) =I2/G2 # 2. Channel Revenue Aggregation (Pivot_Engine Sheet) =SUMIF(Raw_Data!$C$2:$C$101, A2, Raw_Data!$I$2:$I$101) # 3. Safe Division ROAS Formula with Error Handling (Dashboard / Pivot Engine) =IF(D2=0, 0, F2/D2)
```

4. Summary Data Model & Channel Benchmarks

The aggregation engine powers summary tables and top-line executive KPI cards on the dashboard canvas:

Marketing Channel	Total Spend (USD)	Total Revenue (USD)	Blended ROAS	Target ROAS
Google Ads	$25,555.00	$106,911.00	4.18x	3.50x
Meta Ads	$40,320.00	$126,233.00	3.13x	2.80x
LinkedIn	$33,100.00	$153,440.00	4.64x	4.00x
YouTube	$55,180.00	$172,995.00	3.14x	3.00x
Email	$8,295.00	$114,124.00	13.76x	5.00x

5. Troubleshooting & Slicer Interactivity Resolution

During implementation, division-by-zero errors (`#DIV/0!`) in summary ROAS columns were resolved by wrapping formulas in conditional `IF` statements. Slicer interactivity was successfully established by connecting dashboard controls directly to source data ranges and native pivot table arrays, ensuring seamless real-time filtering across all embedded charts.